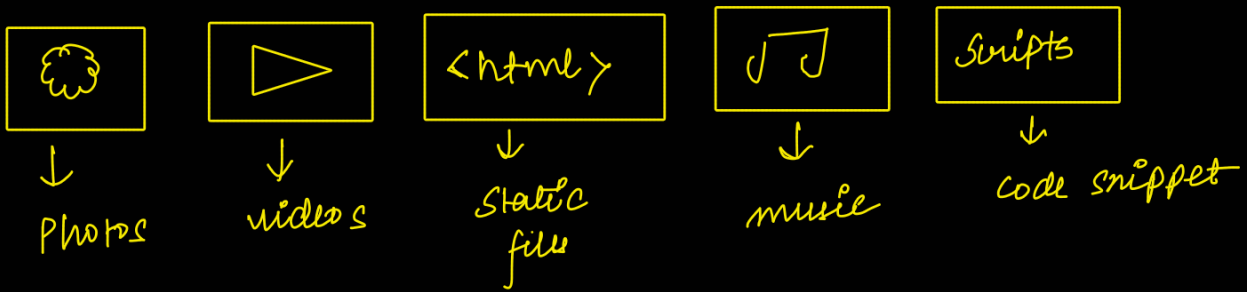
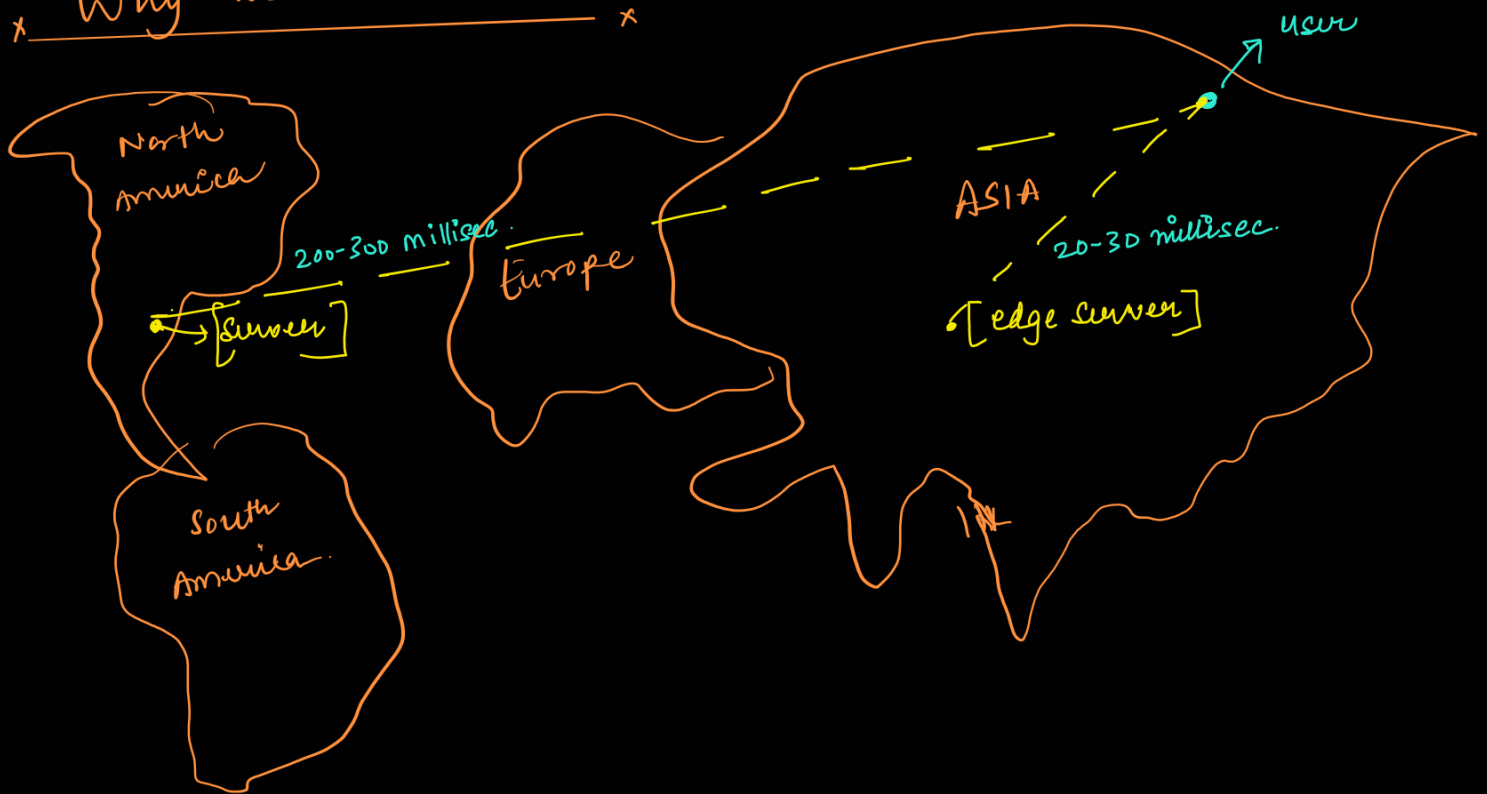


Content Delivery Network



→ to provide large files/static files to users faster and more reliable.

Why we Need CDN



So, CDN are geographically distributed caches for the static content.

Push CDN

① you push the data manually to edge server.

- more Resource utilisation
- optimised for static content
- Reduce load on origin server

Pull CDN

① When user ask for the data it will give (cache miss) and ask for the origin server and next time edge server will have that Data (cache hit) → low support

x TTL x (Time to live)

CDN cache a file for some time,
example [1hr], [1day]

ESI Tag

Utilisation
→ Dynamic content support
→ Scalable

example, Google Cloud CDN, Amazon Cloud front, Akamai

x Content Delivery Network - System Design x

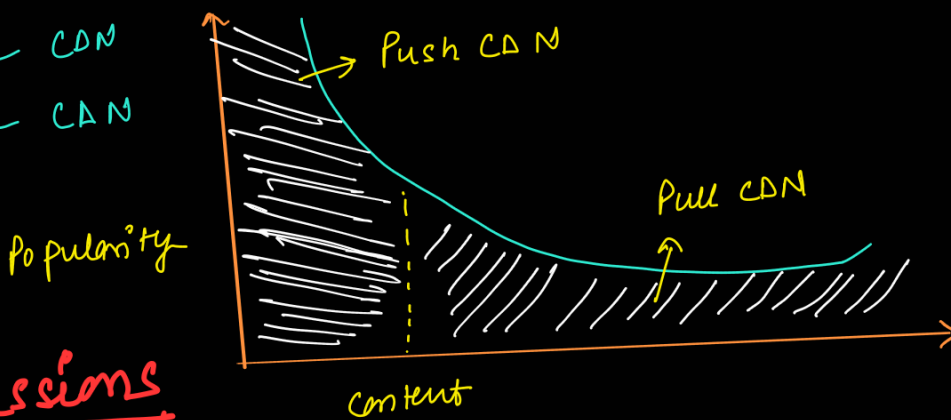
Requirements

- should retrieve data from origin server
- must process the request from end-user
- caching content
- search and locate cached content
- maintain TTL.

- low latency
- high availability
- Global system

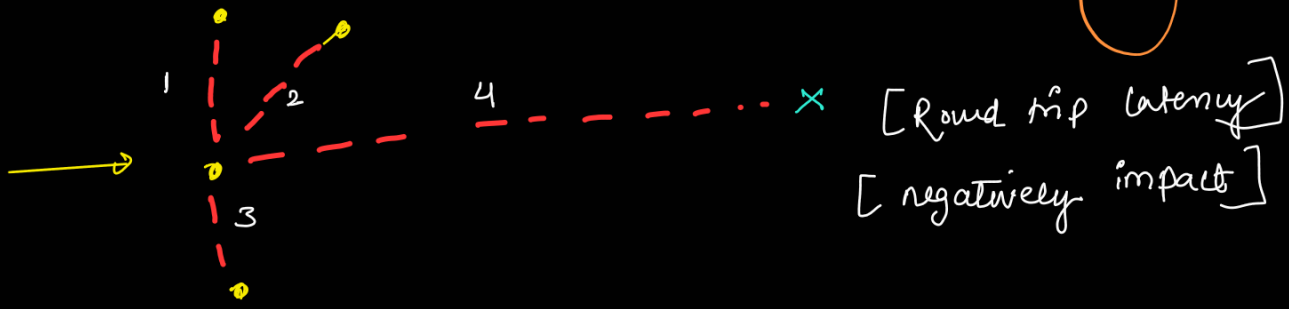
So we know two Caching strategies

- ① Push CDN
- ② Pull CDN

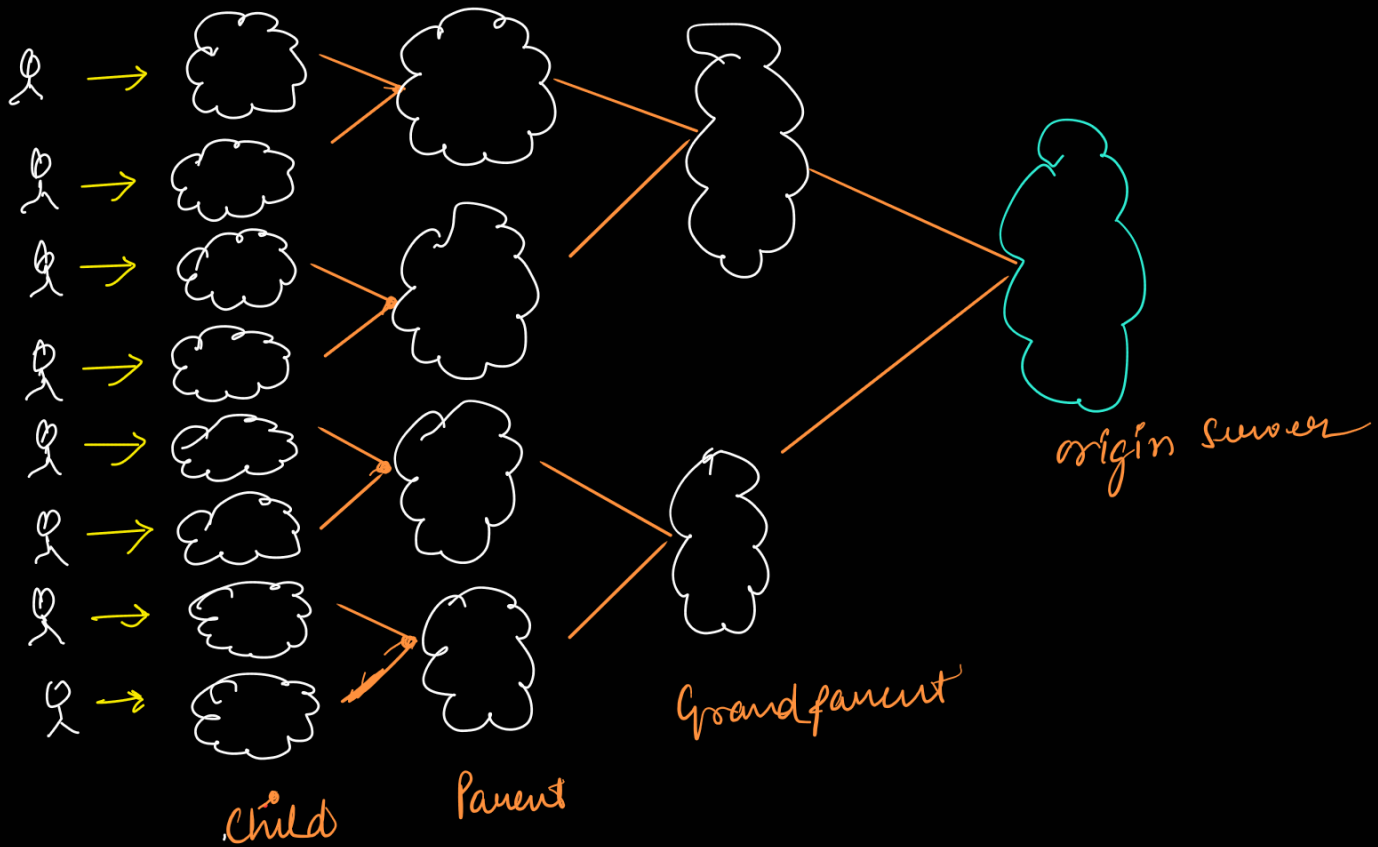


Discussions

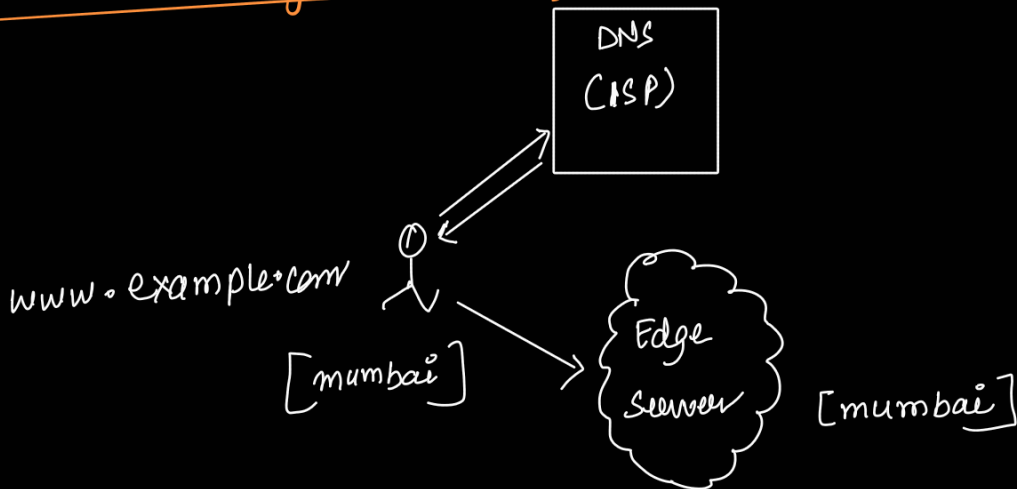




* Multi-Tier CDN *



How to Find Edge Server *



Story

India - edge-ind.cdn.example.com - 132.7.7.58

Germany - edge-ger.cdn.example.com - 194.0.0.1 [London]

www.example.com - 192.10.0.17

New york - edge - NY . com

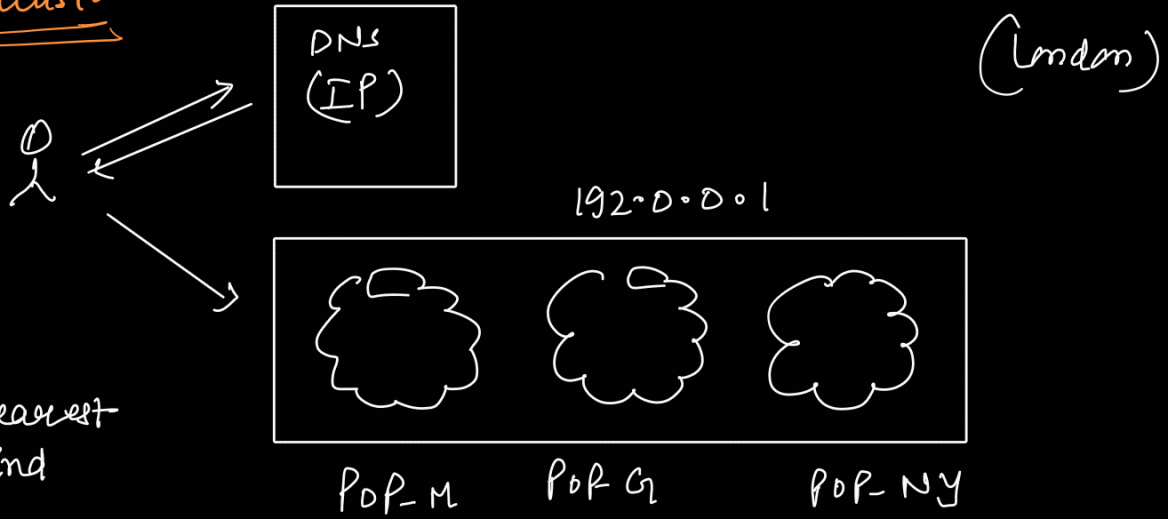
[mumbai] → nearest

Ⓜ → www . example . com

↓ ip [internet service provider] → india

→

AnyCast Cluster



① → nearest find

② → failover

③ → DDoS

④ → load management

Final use case



① how CDN stores Data (key-value)

https://abc.com/index.html

html file

https://abc.com/static/main.js

js file blob

https://abc.com/static/logo.png

image

② how Data is served → Discussed
~ ~ ~ ~ ~

③ who will hit the API?

[edge server], [browser]

→ CDN hit api
and creates the
page and then
serve it

→ client side Rendering

→ CDN used only for static data

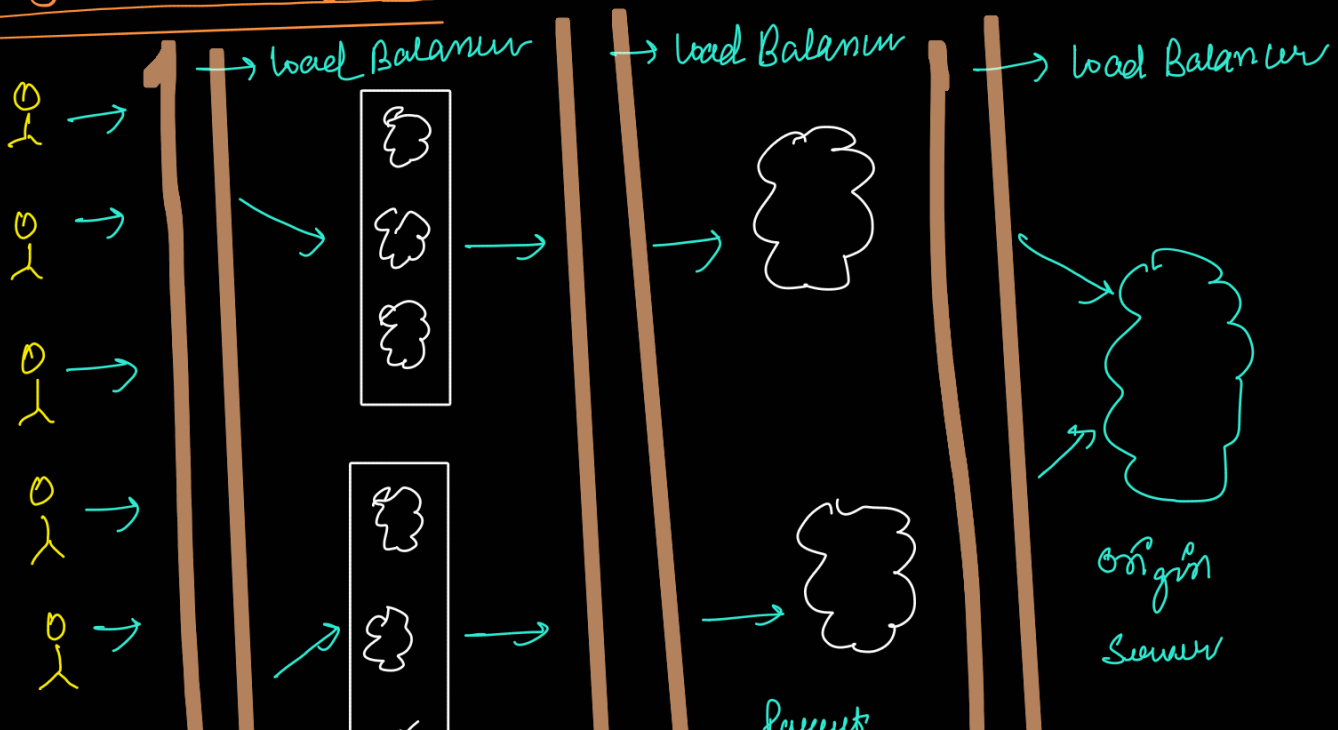
→ Server Side Rendering

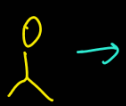
(b) (ESI) → Edge-side includes

↳ so if you have esi-tags call the API

→ and generate html page at server side

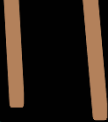
System Design





any
cast

edge server
cluster



primary
server

